

Rolly Fox – Final Project Report

CMS 320: Video Game Design

1. Title, Authors, and GitHub Link

Title: *Rolly Fox*

Authors: Zubair Zabir & Allyson Hussey

GitHub Repository: <https://github.com/ZubairZabir/CMS320-Fall2025>

2. Introduction

a. Background: What is this game about, and why did we choose it?

Rolly Fox is a whimsical, action-based side-scroller inspired by the timeless mechanics of Flappy Bird, but redesigned with a deeply personal twist. Instead of the traditional generic pipes and abstract backgrounds, our game transports players through iconic Rollins College environments, spaces that hold real meaning for students, alumni, and anyone who has spent time on campus. The main character, a small and adventurous fox, symbolizes the spirit of Fox Day: a day filled with joy, unpredictability, and the sense of community that makes Rollins unique. By using the fox as our protagonist, we wanted to capture not only the playfulness of Rollins traditions but also the excitement of navigating college life.

We chose this idea because we felt a strong desire to create something that reflects our own Rollins journey. As we move closer toward graduation, we often find ourselves looking back at the moments, places, and experiences that shaped our time here. Designing Rolly Fox became more than just a class assignment; it became a way to honor those memories in a creative, interactive format. Each level represents a place that has been meaningful to us: from long study nights in Olin Library, to early morning coffee runs at All Saints, to bustling days in the Bush Science Center, and the peaceful sunsets on Lake Virginia. Together, these scenes form a digital scrapbook of our college years.

This project also allowed us to blend creativity, nostalgia, and technical development. We wanted to build something fun and accessible, but also something future Rollins students could enjoy, perhaps even recognizing their own experiences through the game. In many ways, Rolly Fox stands as both a celebration of our time at Rollins and a gift to anyone who shares a connection with the campus. Creating it allowed us to channel our skills into a project rooted in personal meaning, making the game development process more engaging, memorable, and fulfilling.

b. Motivation

Our motivation for creating *Rolly Fox* came from a shared desire to design something that genuinely reflects our time at Rollins College. Over the semesters, we have built countless memories tied to specific places on campus, such as the late-night study sessions, the quiet mornings by the lake, the rush between classes, and the small

moments that make Rollins feel like home. We wanted our final project to capture those experiences in a creative and interactive form, transforming familiar campus locations into an engaging game world.

Another major motivation was accessibility. We wanted to build a game that anyone could pick up and enjoy, regardless of age or gaming experience. By keeping the controls simple and intuitive, using just the space bar, we ensured that players from younger children to older alumni could participate without feeling overwhelmed. The easy-to-learn but hard-to-master gameplay lets casual players enjoy it while still giving competitive players something to improve on.

We were also motivated by the idea of blending humor, nostalgia, and challenge. The fox, representing Fox Day, adds a playful, lighthearted element that instantly connects with Rollins' culture. At the same time, the increasing difficulty of each level mirrors the challenges students face throughout their college journey. This blend creates a game that feels fun, meaningful, and emotionally resonant.

Finally, we wanted the game to serve as a bridge between past, present, and future Rollins students. For current students, the game offers a fun way to see campus from a new perspective. For future students, it acts as a welcoming introduction to the Rollins environment. And for alumni, it becomes a nostalgic reminder of the campus they once called home. Our motivation was not just to complete a class project, but to create something lasting, something that brings people together through shared experiences and campus pride.

c. Target Users

Our target audience includes a wide and diverse range of players who can connect with the game in different ways. First and foremost, *Rolly Fox* is designed for current Rollins students who will immediately recognize the locations, traditions, and atmosphere represented in each level. These players can enjoy the game not only for its mechanics but also for the nostalgia and familiarity woven into the environment. Former students and alumni are also a major part of our target audience, as the game offers them a playful way to revisit places they once spent years studying, relaxing, and building community. In many ways, the game becomes a digital way to “walk through campus” regardless of where they are in the world.

We also considered prospective students who may be exploring Rollins College for the first time. For them, the game serves as an interactive introduction to campus culture. Instead of simply seeing the campus on a website or tour, they can engage with it through a fun and welcoming medium that highlights key locations and the spirit of Fox Day. This makes *Rolly Fox* both an entertaining experience and a unique way to showcase Rollins to future students.

Faculty and staff who work on campus are another important group. Many of them spend years in the same buildings and may enjoy seeing these familiar spaces depicted in a stylized, lighthearted way. The game allows them to experience campus from a fresh perspective and appreciate the creativity behind transforming everyday environments into playful game levels.

Finally, we intentionally included children and casual players as part of our audience. Because the game uses a single control, tapping the space bar, it is easy for younger players or those unfamiliar with gaming to jump in quickly without needing to learn complex mechanics. At the same time, the progressive

difficulty offers enough challenge to keep adults engaged, especially those who enjoy trying to beat their own high scores. The accessibility of the controls makes the game approachable for players roughly between the ages of 8 and 65, ensuring that it remains fun, inclusive, and enjoyable for all experience levels.

d. Game Context / World

The world of *Rolly Fox* consists of four Rollins-themed levels:

1. **All Saints Café** – A cozy, familiar hangout spot for students
2. **Bush Science Center** – Represents the academic rigor of STEM students
3. **Olin Library** – The heart of studying and exam preparation
4. **Lake Virginia** – Symbolizes peace, reflection, and campus beauty

Each scene contains unique obstacles, colors, and environmental details that match the theme.

e. Game Rules

- Tap **Space** to keep Rolly (the fox) in the air
- Avoid all obstacles
- Hitting anything results in an instant game-over
- Each cleared obstacle = **+1 point**
- Speed and difficulty varies in each scene
- Goal: beat your previous high score

f. Controls

- **Keyboard:** Press *Space* to flap / jump

g. Rough Sketches

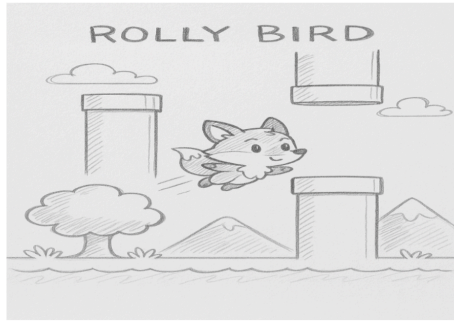
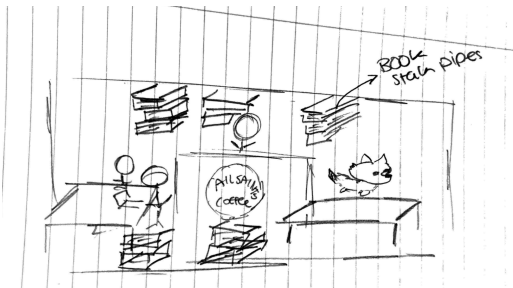


Fig: Sketch of Rolly Fox

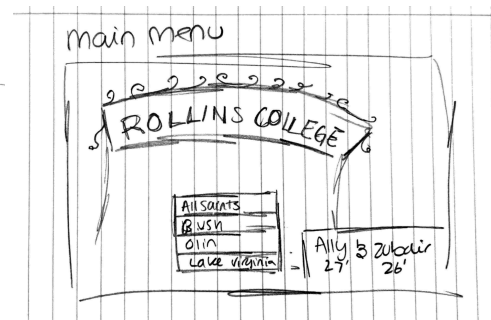


Fig: Sketch of Game Over

Ally: Draft of All Saints level



Ally: Menu layout



h. Task Distribution Among Teammates

Zubair

- Game programming and core mechanics
- Scene switching and level loading
- Collision detection and scoring logic
- Creating prototypes of all four scenes
- Bug fixes, balancing, and quality assurance

Ally

- Game art and environment design
- Level-specific pipe designs
- Adjusting physics for different levels

- Creating consistent color palettes and aesthetic
- Visual polish and UI adjustments

Shared Responsibilities

- Playtesting
- Difficulty balancing
- Final polish and presentation

3. Game Development

We approached the development of *Rolly Fox* in stages that followed the order of how gameplay unfolds.

Base of the Game

We started with the core mechanic: *Flappy Bird-style physics*.
This included:

- A rigidbody for the fox
- Gravity settings
- Upward impulse on key press
- Procedural obstacle spawning
- Collision detection and score tracking
- Creating a main menu scene

Once the base worked smoothly, we expanded the game into a multi-scene project.

Scene 1: All Saints Café

- Designed as the introduction level, easy difficulty
- Warm colors and familiar café background
- Simple pipe spacing to let players adjust to the controls

- Ally designed the All Saints textures
- Zubair implemented the first version of obstacle movement & spawn timing

Scene 2: Bush Science Center

- Increased speed and slightly tighter pipe gaps
- Science-themed obstacles matching the academic theme
- Background includes lab silhouettes, science posters, and blue hues
- Physics adjusted by Ally to feel “more intense,” matching the building’s energy

Scene 3: Olin Library

- Designed to feel “quiet but stressful”
- Pipes themed like bookshelves and furniture
- Lighting adjusted to look dimmer and more focused
- Speed increases gradually over time
- Obstacles spawn faster to simulate finals season pressure

Scene 4: Lake Virginia

- Final and most peaceful level
- Background with calm water, boats, trees
- Soft gradients to mimic sunset over the lake
- Slow start but sharp difficulty spike
- Zubair implemented the variable difficulty curve
- Ally created the water and shoreline textures

Final Provisions

Before submission, we implemented:

- A unified main menu that links all four scenes
- Restart and quit buttons
- Polished animations for Rolly
- Score UI and high score logic
- Bug fixes (e.g., infinite falling, collision glitches)
- Audio adjustments

4. Final Game Video

Gameplay Recording:

<https://github.com/ZubairZabir/CMS320-Fall2025/blob/main/Gameplay.mp4>

5. Discussion and Future Work

How We Felt Developing the Game

Developing *Rolly Fox* was a genuinely meaningful experience for both of us, not only because it required technical and artistic creativity, but because it allowed us to transform our personal Rollins memories into something interactive and playable. Each level we designed brought back specific emotions tied to campus life, working on All Saints Café reminded us of slow mornings spent grabbing coffee before class, and adding details to the Bush Science Center reminded us of late nights before exams and the energy of the academic environment. The Olin Library scene carried a sense of calm focus, reflecting the many hours students spend preparing for tests or writing papers. Finally, building the Lake Virginia level was peaceful and nostalgic, reminding us of sunsets, walks by the water, and some of the most serene moments of our college journey.

The process also helped us appreciate our individual strengths and how they complemented each other. Zubair focused on programming, mechanics, and technical problem-solving, while Ally shaped the art direction, aesthetics, and environmental design. This division of skills allowed the project to grow quickly and cohesively. Collaborating on playtesting, balancing the difficulty, and polishing the final version brought us even closer to creating something we're proud of. In many ways, completing this game felt like a celebration of our teamwork and our shared experiences at Rollins.

Challenges and How We Overcame Them

1. Resolution and Window Mode Issues

One of the earliest challenges we faced involved exporting the game properly on macOS. The builds often opened in full-screen mode with distorted resolution, which made it difficult to see or control the game. This became frustrating because every time we fixed one visual issue, another would appear in the build settings.

How we fixed it:

We experimented with Unity's build configuration, manually adjusting resolution presets, changing the aspect ratio, and enabling the correct windowed mode options. After multiple test builds, we found a stable configuration that displayed consistently across macOS devices.

2. Physics Inconsistencies Between Scenes

Because each level had different backgrounds, obstacles, and visual density, they initially felt disconnected from one another. The fox would float smoothly in one scene but feel heavier or more erratic in another, creating an uneven gameplay experience.

How we fixed it:

We standardized the Rigidbody and gravity settings across all scenes and then fine-tuned the obstacle spacing, pipe speed, and difficulty curve to ensure each environment felt challenging but fair. This brought a consistent "feel" to the entire game.

3. Collision Bugs

Throughout testing, we noticed several instances where obstacles failed to register collisions. This caused players to pass straight through pipes without triggering a game-over, making the game unpredictable and unbalanced.

How we fixed it:

We reviewed all collider components across the obstacles and expanded their collision boxes to ensure proper overlap. We also fixed the Unity layer interactions so that the fox and obstacles were properly assigned to detectable layers. These adjustments ensured reliable collision detection across all four scenes.

4. Team Scheduling

As with many group projects, aligning schedules was occasionally difficult. We each had different academic commitments, exams, and external responsibilities, making it challenging to sit down together for long work sessions.

How we fixed it:

We divided responsibilities according to our weekly availability and communicated frequently through text and shared notes. This allowed us to keep the project moving smoothly even when we couldn't meet in person. By setting small goals each week, we kept the game progressing at a steady pace while avoiding last-minute stress.

Feedback during Game Showcasing

Throughout the event, students, faculty, alumni, and family members stopped by to play the game and share their feedback. It was refreshing to see players genuinely enjoying our creation. We received only one piece of negative feedback, which focused on the birds in the final scene at Lake Virginia. They were considered "not fitting the aesthetic," and we agree that the bird design can be improved to better match the overall vibe. Beyond that, many participants described the game as great, competitive, engaging, and colorful. We also received numerous comments praising how realistic and accurate the features were to Rollins. This was especially rewarding to hear, as it aligned with several of our main project goals. We wanted players to feel connected and emotionally relate their experience back to Rollins, and overall, the game was well-received.

Future Work

Looking ahead, there are several exciting features we would like to add if we continue developing *Rolly Fox*. These include:

- **Animated NPCs**, such as students walking or boats moving across Lake Virginia
- **Power-ups**, including shields or slow-motion bonuses
- **A global leaderboard system** that tracks high scores across players
- **Seasonal skins** for the fox (e.g., Fox Day sunglasses, graduation cap)
- **A hidden fifth level**, unlocked by completing the first four scenes
- **More advanced difficulty scaling**, allowing players to progress beyond the base challenge

Our goal is for *Rolly Fox* to continue evolving into a more complete and polished game, one that players can return to repeatedly while enjoying new features and stronger connections to the Rollins campus.